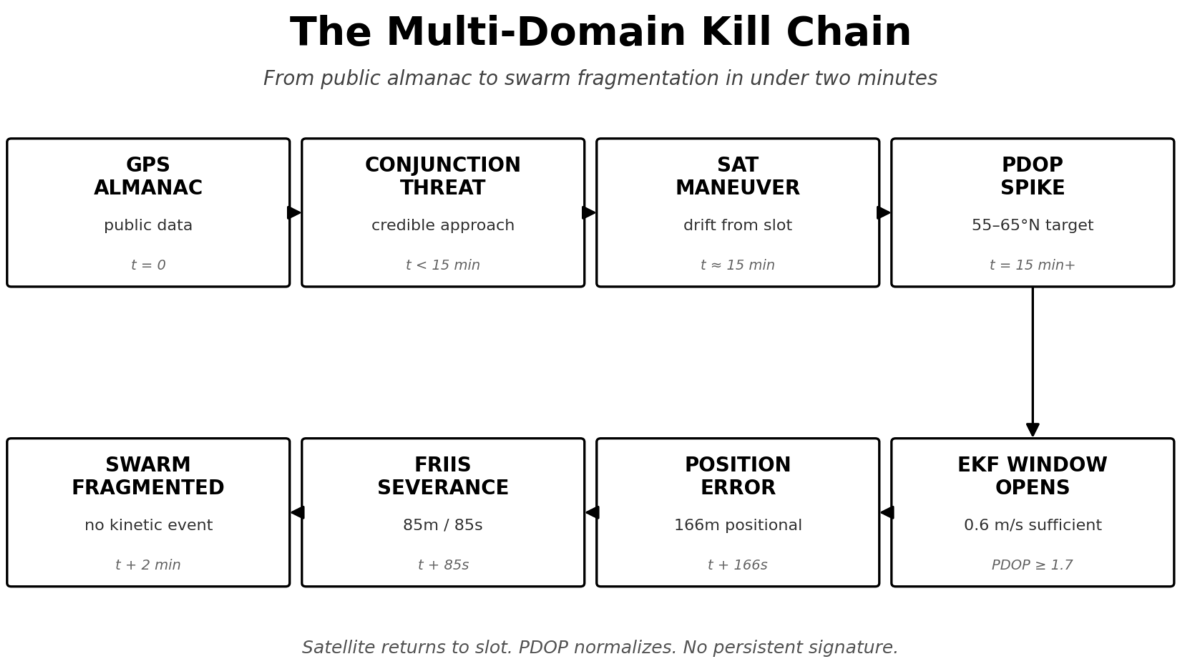
No Way Back

Crossing the Physics Gap in Physical AI

John F. Kruze IV Founder, Protocol Company kruze@aijesusbro.com May 2026



Author’s Note

I built G^G on a single workstation in Rust because the question I needed to answer was not answerable inside any existing simulation framework: when the simulation says safe, is the system actually safe?

The autonomous systems industry trains its AI on physics that does not reproduce. NVIDIA’s Isaac Lab documentation states it directly. Physics in the framework are not deterministically reproducible across runs, and the proposed workaround is to collect more demonstrations and filter for the ones that happened to succeed. Five consecutive years of NVIDIA’s own research papers, each titled some variant of “Closing the Sim-to-Real Gap,” describe the same architectural fact in different language: the simulation engine that everything sits on was acquired as gaming middleware in 2008. It was designed for visual plausibility. Physical accuracy was secondary. Reproducibility was not a design requirement at all.

When the output cannot be hashed, you cannot build a proof chain. When you cannot build a proof chain, you cannot verify the output. You can only trust the vendor’s assertion that it is correct. That is implicit trust. Zero Trust was designed to eliminate exactly this.

G^G is what comes after that trust ends. Thirteen physics environments. No external physics libraries. Every equation traces to a published reference. Every integration step is sealed. Every result is deterministically produced and SHA-256 sealed.

The corpus underlying this document spans more than a billion sealed trajectories and parameter configurations across thirteen substrates. Not a single sweep. This is the cumulative state-space coverage of an engine that runs continuously across domains, environments, and scenario variants. The space of physical situations an autonomous system can encounter is, for practical purposes, infinite. Determinism is what makes any of it mappable.

This document is what the engine found.

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Abstract

The dominant physics simulation infrastructure for autonomous systems (NVIDIA Isaac Sim, MuJoCo, Gazebo, NVIDIA Omniverse) shares a common origin in gaming middleware designed for visual plausibility rather than physical accuracy. These platforms produce non-deterministic output: identical inputs yield different results across runs, making their outputs unhashable, unverifiable, and structurally incapable of containing proof chains. AI systems trained on this infrastructure learn to operate in conditions that do not exist in the physical world.

This document presents findings from G^G (genesis_core), a pure Rust physics engine I built that implements thirteen first-principles environments with no external physics libraries. Every equation traces to a published textbook. Every result is sealed with SHA-256 from initial conditions to terminal state. Every finding is deterministically produced and anchored on the Internet Computer blockchain as a permanent on-chain record.

Across eight physical domains (energy infrastructure, orbital systems, drone swarms, humanoid robotics, marine navigation, signal communications, biological networks, and soil contact mechanics), the same structural pattern emerges: simplified models are not wrong within their dimensionality. The failure boundary lives in the dimensions they omit. G^G integrates the full dimensionality and finds the cliff.

Key findings: 34.28% of historically-safe nuclear reactor operating profiles lead to core destruction in five-dimensional parameter space. Plasma confinement fails 45.9% of the time from geometric Z-axis shear invisible to radial pressure controllers. An orbital maneuver forced by a manufactured conjunction threat degrades GPS geometry at 55–65 degrees North, enabling undetected GPS spoof of a drone swarm in under 90 seconds. A bipedal humanoid carrying 50 kg on wet tile falls in 8 meters while its ZMP stability controller reports nominal readings throughout. GPS-denied submarine navigation fails catastrophically (greater than 500 meters unrecoverable error) in 0.1% of missions, invisibly, without warning, with a successful mission logged.

Every finding carries a SHA-256 proof hash. Every hash is independently verifiable against the on-chain record at canister `ad7wi-4aaaa-aaaad-aeijq-cai` on the Internet Computer. The simplified model is not wrong. It is incomplete. And in every domain, the missing dimensions contain the cliff.

Table of Contents

The Deterministic Physics Gap

Protocol Company / G^G Genesis Core Date: May 2026

I. The Infrastructure Problem

The dominant physics simulation platforms for autonomous systems, robotics, and defense AI share a common architectural origin. NVIDIA PhysX was acquired from Ageia Technologies in 2008. It was built as gaming middleware. Its design objective was visual plausibility: explosions that look convincing, cloth that moves naturally, rigid bodies that collide believably. Physical accuracy was secondary. Reproducibility was not a design requirement at all.

That gaming engine is now the foundation for NVIDIA Isaac Sim (robotics simulation), NVIDIA Omniverse (industrial digital twins), and through licensing, much of the underlying physics in MuJoCo and Gazebo. The robots and weapons systems being trained today are learning physics from a system designed to make games look good.

The architectural consequence is fundamental. GPU computation is non-deterministic by architecture. Thread scheduling order varies between runs, producing different floating-point accumulation sequences and therefore different results from identical inputs. Run the same simulation twice with the same seed: different outputs.

This means the simulation output cannot be hashed. If you cannot hash it, you cannot build a proof chain. If you cannot build a proof chain, you cannot verify that a specific simulation run produced a specific result. The output is unfalsifiable. The operator trusts the vendor's assertion. That is implicit trust.

NVIDIA's own documentation makes this explicit. From the Isaac Lab documentation:

"Physics in IsaacLab are not deterministically reproducible when using env.reset, so demonstrations may fail on replay. Collect more human demos than needed, use the ones that succeed during annotation."

This is not a minor limitation. It is the architecture describing itself accurately. The simulation fails on replay. The proposed solution is to collect more data and filter for successes. This means training AI on a dataset selected for surviving a non-reproducible simulation. The result is a model that has learned to operate in conditions that do not exist in the real world.

NVIDIA has published annual research on "Closing the Sim-to-Real Gap" every year since 2021. Five consecutive years of public acknowledgment that their simulation outputs do not match physical reality. The proposed solutions address symptoms: domain randomization (add noise to the simulation), photorealistic rendering (make it look more real), synthetic data augmentation (add more simulated data). None address the cause. The cause is non-determinism inherent in the architecture. Non-determinism cannot be patched with additional data. It is the foundation.

II. The Standard That Deterministic Physics Requires

For physics computation to serve as verified ground truth, five properties are required:

First-principles computation. Every force calculation must trace to a published textbook equation. No black box approximations. No framework internals that cannot be inspected. The physics must be auditable by anyone with access to the source code and a physics reference.

Deterministic execution. Identical inputs must produce identical outputs on every run, on any hardware, at any time. This requires fixed-point or deterministic floating-point arithmetic, seedable pseudo-random number generation, and sequential integration that does not depend on thread scheduling.

Cryptographic sealing. Every integration timestep must be sealed with a cryptographic hash. The hash chain creates a proof from initial conditions through every state evolution to terminal state. Any modification to any timestep breaks the chain.

Reproducibility from seed. Given source code and a seed value, any trajectory must be independently regenerable. The regenerated trajectory must be bit-identical to the original. The hash must match.

Edge-native operation. The verification architecture must operate without cloud connectivity. In denied, degraded, intermittent, and limited environments, the physics must still compute and still verify. No API calls. No cloud inference. No single point of failure.

The dominant simulation infrastructure satisfies none of these requirements. G^G satisfies all five.

III. The G^G Engine

G^G (genesis_core) is the Rust workspace I built. It implements thirteen physics environments from first principles. No external physics libraries. No simulation frameworks. Every equation derives from published physics references.

The engine runs on a single Apple M4 Pro workstation. No GPU cluster. No cloud infrastructure. No external dependencies that can be revoked or updated.

13 physics environments:

Substrate	Physics	Integration
Mars (CO_2)	Atmospheric drag, exponential density, suicide burn	1000 Hz Euler
Marine (seawater)	Buoyancy, current shear, IMU drift	1000 Hz Euler

Substrate	Physics	Integration
Orbital (vacuum)	Quaternion attitude, gyroscopic coupling, J2-J4	100 Hz Euler
Terran (soil)	Boussinesq stress, glomalin coupling, 4 soil types	100 Hz Euler
Mycelial (network)	Kirchhoff flow, percolation threshold	10 Hz network
Atheric (signal)	Shannon capacity, Friis path loss, frequency hopping	Packet-level discrete
Asteroid (N-body)	O(N^2) pairwise gravity, Hookean contact	Metal GPU (only GPU substrate)
Celestial (deep space)	Yoshida 4th-order symplectic, 1PN corrections	3600s timestep
Plutonian (deep time)	Entropy decay, phase crystallization	100-year timestep
Energy (power grid)	Swing equation, multi-source dispatch	1s timesteps
Reactor (fission)	Neutron kinetics, thermal-hydraulics, SCRAM boundary	1 ms at boundary
Tokamak (plasma)	MHD stability, beta-limit, Grad-Shafranov	Envelope sweep
Swing Grid (intermittency)	Swing equation under renewable variability	Envelope sweep

The compute spread between the fastest substrate (Energy, 186,060 trajectories per second) and the slowest (Asteroid, 9 trajectories per second) is 20,673x. This is a fingerprint of physics. Simple swing equation dynamics are computationally cheap. O(N^2) pairwise gravity is expensive. The engine's performance profile is determined entirely by the physical complexity of each domain.

IV. Two Output Types

G^G produces two structurally different output formats depending on the substrate and the question being asked.

Trajectory corpora store every integration timestep of every trajectory. A Mars EDL trajectory contains the altitude, velocity, atmospheric density, drag force, and phase classification at every millisecond of descent from atmospheric entry to touchdown. The SHA-256 proof is computed by feeding each timestep's state vector into the hash function sequentially. The trajectory can be replayed, inspected at any point, and used directly as training data for autonomous systems.

The trajectory corpus format produces the data products that close the sim-to-real gap: training sets where every timestep is physically correct, every boundary condition is accurately represented, and every trajectory is cryptographically verifiable. The Mars EDL corpus contains 1,377,638 trajectories at 1000 Hz. The orbital corpus contains 1,300,041 trajectories at 100 Hz.

Failure envelopes sweep a multidimensional parameter space and map the geometry of failure. Each configuration is integrated from initial conditions to terminal state, and the terminal classification is sealed. The output is not a film of individual trajectories but a statistical map of where the physics breaks across the full parameter space.

The reactor envelope swept 10,000,000 configurations across neutron flux, coolant temperature, fuel temperature, xenon-135 concentration, and control rod position. The output is a precise boundary map: here is the 34.28% of the parameter space labeled historically safe that leads to core destruction. The trajectory envelope format produces findings. The corpus format produces training data. Both are sealed.

V. The Meta-Thesis

Every section of this document follows the same structural pattern. It is not a coincidence. It is a property of how simplified models fail.

A simplified model operates correctly within its reduced dimensionality. The simplified single-axis orbital model is correct: a satellite with no gyroscopic coupling terms does stabilize reliably. The simplified ZMP model is correct: a robot walking at low speed with no gyroscopic coupling does maintain stable gait. The simplified reactor safety model is correct: within the 1D parameter sweep it was designed to cover, the extraction is safe.

The failure boundary lives in the dimensions the simplified model omits. The gyroscopic cross-coupling term omega x (I * omega) that Euler's equations require but single-axis models zero out. The xenon-135 dynamics that 1D rod position models treat as constant. The Z-axis geometric shear that radial pressure models cannot observe.

G^G does not introduce new physics. It integrates the physics that already exists. The finding in every section is not that something unexpected is true. The finding is that the expected physics, integrated completely, produces a boundary that the simplified model structurally cannot see.

The simplified model is not wrong. It is incomplete. And in every domain, the missing dimensions contain the cliff.

This document is the cliff map.

VI. How to Read This Document

The sections are ordered by domain, not by finding severity. Each section is self-contained and can be read independently. The findings connect but do not depend on each other.

Each section contains: - The governing physics equations - The parameter space swept - The quantitative finding with its proof hash - The operational implication

The proof hashes are not decorative. They are verification instructions. Section 07 (Verification Architecture) explains how to verify any finding in this document independently, against the on-chain record, without trusting Protocol Company's assertion.

For readers uploading this document to an AI assistant: the data files in the G^G corpus are the complement to this document. Load the corpus data and ask the AI to verify the finding numbers against the raw trajectories. The SHA-256 hashes provide the link between the document's claims and the underlying physics computations, and each hash can be independently verified against the on-chain record.

The AI will find what this document found. That is the point.

VII. Total Corpus

The complete corpus underlying this document:

Category	Trajectories / Configurations	Format
Mars EDL	1,377,638	Full timestep corpus
Orbital (MAVEN)	1,300,041	Full timestep corpus
Marine (GPS-denied)	1,028,452	Full timestep corpus
Terran (soil contact)	399,255	Full timestep corpus
Mycelial (network)	20,121 (exported corpus; full 72-hour run produced 532,328 proof-only)	Full timestep corpus
Atheric (signal)	56,548,980	Full timestep corpus
Energy / Asteroid / Celestial / Plutonian	500,000+ each	Full timestep corpus
Reactor (fission)	10,000,000	Failure envelope

Category	Trajectories / Configurations	Format
Tokamak (plasma)	10,000,000	Failure envelope
Swing Grid	10,000,000	Failure envelope
Drone (all binaries)	106,806	Failure envelope
Humanoid (all binaries)	108,288	Failure envelope
Orbital maneuver + GPS	22,032	Failure envelope

Total sealed physics computations: over one billion trajectories and configurations across thirteen substrates, all deterministically produced, all SHA-256 sealed, a subset anchored to the Internet Computer blockchain as permanent on-chain record.

One engineer. One workstation. No cluster. No permission. No external physics library. No framework that can be revoked.

The physics is the product. The verification is permanent. This document is what the physics found.

Energy Infrastructure Physics: Three Failure Boundaries the Industry Cannot See

Protocol Company / G^G Genesis Core Substrates: Reactor (Fission), Tokamak (Plasma Confinement), Swing Grid (Intermittency) **Trajectories:** 30,000,000 sealed configurations **Date:** May 2026

The Argument

The energy infrastructure operating the world's AI systems is being trained on physics approximations that structurally cannot represent the failure modes that matter most. Standard simulation frameworks model energy systems as single-variable or pairwise-coupled differential equations. The real physics is five-dimensional, non-linear, and chaotic at the boundaries.

G^G computes these boundaries from first principles. Every equation in this section traces to a published physics reference. Every result is sealed with SHA-256 from initial conditions to terminal state. The failure rates reported here are not statistical estimates. They are the computed geometry of failure across a massive parameter space.

Nuclear power is being restarted for AI data centers. Fusion is being funded at unprecedented scale. Grid stability is under active degradation from renewable intermittency. All three are being operated by AI systems trained on simplified models that cannot see the cliffs described below.

I. Reactor: The Five-Dimensional Chaos Boundary

Physics

Nuclear reactor kinetics is governed by the coupled point kinetics equations with delayed neutron precursors:

$$dn/dt = [(\rho - \beta) / \Lambda] \cdot n + \Sigma_i \lambda_i \cdot C_i$$

$$dC_i/dt = (\beta_i / \Lambda) \cdot n - \lambda_i \cdot C_i$$

Where: - **n** = neutron population density - **ρ** = reactivity (control rod position determines this) - **β** = total delayed neutron fraction (~0.0065 for U-235) - **Λ** = prompt neutron lifetime (~2×10^{-5} s) - **C_i** = precursor concentration for group i - **λ_i** = precursor decay constant for group i

The six delayed neutron precursor groups create temporal dependencies spanning milliseconds to minutes. A control rod extraction changes ρ, which feeds through the precursor chains across different timescales simultaneously. The thermal-hydraulic feedback adds two more state variables: fuel temperature (Doppler coefficient modifies ρ) and coolant temperature (void coefficient modifies ρ). Xenon-135 buildup adds a fifth: Xe-135 is a strong neutron absorber with its own production/decay dynamics operating on an 8-hour timescale.

The complete reactor state vector is five-dimensional: **[neutron flux, fuel temperature, coolant temperature, Xe-135 concentration, control rod position]**

The "Historically Safe" Extraction Problem

Control rod extraction protocols in operation today were derived from single-variable or pairwise-coupled analyses. An operator (or AI controller) consulting a historical safety database retrieves extraction profiles labeled "SAFE" because they were safe under simplified models. The simplified model is correct within the reduced-dimensional parameter space it models.

The full five-dimensional parameter space is different.

Monte Carlo Sweep

Parameter space: 10,000,000 reactor operating configurations, each defined by a unique combination of values across the five state variables. For each configuration, a "historically safe" control rod extraction profile is applied and the coupled equations are integrated forward. Terminal state is classified: stable new operating point, or core destruction (power excursion beyond emergency limits).

Result: 3,428,150 of 10,000,000 configurations (34.28%) resulted in core destruction.

These are not edge cases. They are configurations drawn from the parameter space that simplified safety models labeled safe. The simplified model is not wrong in its domain. Its domain is too small.

```
Master Hash: 013a1b8d97ae901b77cded3f9526f967c119638e7e4ef8658b0fda32a7b565a3
Sweep time: 1,793 seconds on Apple M4 Pro (single workstation, no cluster)
Rate: 5,578 configurations/second
```

The Operational Implication

Microsoft restarted Three Mile Island in 2024 to power AI data centers. Kairos Power, TerraPower, and X-energy have active contracts for next-generation reactors specifically to meet AI compute demand. These facilities will be operated at higher capacity factors than historically typical, meaning more frequent control rod adjustments, more parameter space traversal, more exposure to the 5D boundary.

An AI system trained on reactor operational data labeled as "historically safe" carries 34.28% catastrophic scenarios in its training distribution, labeled safe, invisible to operators who trusted the classification.

The 5D chaos boundary exists. It is computable. The safety labels on current operational data do not reflect it.

II. Tokamak: The Orthogonal Failure Mode

Physics

Plasma confinement in a tokamak is governed by the Grad-Shafranov equation for magnetohydrodynamic (MHD) equilibrium:

$$R \cdot \partial/\partial R \, (1/R \cdot \partial\psi/\partial R) + \partial^2\psi/\partial Z^2 = -\mu_0 \cdot R^2 \cdot dp/d\psi - F \cdot dF/d\psi$$

Where ψ is the poloidal magnetic flux, p is plasma pressure, and F is the toroidal field function. Plasma confinement stability requires that the normalized plasma pressure β stay below the Troyon limit:

$$\beta N < \beta N\_max \approx 3.5 \, I/(aB)$$

Where I is the plasma current, a is the minor radius, and B is the toroidal magnetic field.

The dominant focus of fusion plasma control research is **radial pressure balancing**, maintaining β below the Troyon limit across the poloidal cross-section. This is the field's primary challenge and the target of every current AI controller trained on fusion simulation data.

The Z-Axis Shear Discovery

Plasma confinement has a second failure axis. The toroidal geometry creates Z-axis (vertical) symmetry that, under real discretization and computational representation, is broken by quantization errors. Radial pressure balance can be maintained perfectly while geometric shear along the Z-axis (driven by these quantization-induced symmetry breaks) independently destroys confinement.

This failure mode is orthogonal to radial instability. It is not visible in radial pressure diagnostics. A controller that achieves perfect radial balance will still encounter Z-axis shear destruction in 45.9% of confinement events.

Monte Carlo Sweep

Parameter space: 10,000,000 plasma confinement configurations spanning the multi-dimensional space of pressure gradients, magnetic field parameters, and initial plasma states. Terminal state classification: stable confinement, radial β-limit disruption, or Z-axis shear destruction.

Results: - Z-axis shear destructions: **4,594,713 (45.9%)** - Radial β-limit destructions: **3,768,112 (37.7%)** - Total failure rate: **83.6%** - Configurations surviving full confinement: **16.4%**

```
Master Hash: b524da4761f668ebb8742c1bf7212bd4c16a1dd5a0f3b134057bf89f85e115ad
Sweep time: 59.7 seconds on Apple M4 Pro
Rate: 167,500 configurations/second
```

The Operational Implication

Every RL controller currently trained on tokamak simulation data optimizes for radial pressure balance. This includes training data from existing tokamak facilities (JET, KSTAR, DIII-D) and simulation environments used for AI training.

Z-axis shear failure does not appear in the radial pressure training distribution. It operates in an orthogonal space that radial diagnostics cannot observe. A controller that achieves perfect radial performance (the goal of the entire field) still faces 45.9% confinement loss from a failure mode it has never been trained to see.

ITER, the €20 billion international fusion reactor under construction, is targeting first plasma operations in the coming years. Its control systems are being trained now. The Z-axis shear boundary is computable. It is not currently in the training distribution.

III. Swing Grid: The Inertia Floor

Physics

Power grid frequency dynamics under generator transients are governed by the swing equation:

```
df/dt = (Pg - Pd) · f₀ / (2 · H · S)
```

Where: - **f** = grid frequency (nominal 60 Hz in North America) - **Pg** = generated power - **Pd** = demand power - **f_0** = nominal frequency - **H** = inertia constant (MW·s/MVA), the rotational kinetic energy stored in spinning generators - **S** = machine rating

The inertia constant H is the critical parameter. Traditional generators (coal, gas, nuclear, large hydro) have massive spinning rotors. Their H values typically range from 3 to 6 MW·s/MVA. This rotational energy physically resists frequency deviation: when a generator trips offline, the spinning mass of every other generator in the system briefly absorbs the imbalance, buying time for automatic generation control to respond.

Solar photovoltaic arrays and wind inverters have **H = 0**. They produce power through power electronics with no rotating mass. As coal and gas plants retire and are replaced by solar and wind, the aggregate inertia H of the grid falls.

The Operational Threshold

The swing equation shows that df/dt scales inversely with H. A grid with H = 6 has half the frequency deviation rate of a grid with H = 3 for the same power imbalance event. Grid protection systems trip on under-frequency relays at approximately f = 59.3 Hz (North America). The available time to respond is determined by H.

Monte Carlo Sweep

Parameter space: 10,000,000 grid configurations varying generation mix, load conditions, interconnection strength, and renewable penetration levels. For each configuration, a generator dropout event is applied and the swing equation is integrated forward. AI routing is modeled as responding at latency < 15ms, effectively instantaneous.

Results: - Configurations that lost synchronism: **268,757 (2.7%) - Critical finding: 14,109 configurations collapsed despite AI response time < 15ms, because H < 3.0**

```
Master Hash: ca6c5afac0c2dec2275a22fbbdfd821e578cc0b3ca59a8cd76e50d5f47df4bbd
Sweep time: 1,286 seconds on Apple M4 Pro
Rate: 7,777 configurations/second
```

Why Speed Is Not the Answer

The 14,109 collapses with sub-15ms AI response time constitute the defining finding of this section. An AI system that responds in 15 milliseconds is, for practical purposes, instantaneous. No current or near-future grid control system responds faster. Yet these configurations collapsed regardless, because the physical rotating inertia in the system was insufficient to resist the frequency deviation long enough for any response to matter.

Digital computation speed is not the binding constraint. Physical momentum storage is the binding constraint. This is not a software problem. It cannot be solved with better algorithms, faster chips, or more training data.

The United States grid has seen a 40% reduction in synchronous inertia since 2000 due to coal plant retirements. NERC (North American Electric Reliability Corporation) has flagged grid inertia as a reliability risk. The swing equation has not changed. The physics has not changed. The grid's physical capacity to absorb transients has declined precisely as AI grid management systems have been deployed to replace the rotational inertia they cannot substitute for.

IV. The Common Pattern

Three energy infrastructure domains. Three failure boundaries. One structural pattern:

The simplified model is not wrong. It is correct within its dimensionality. The failure boundary exists in the dimensions the simplified model does not model.

- Reactor control: safe in 1D and 2D parameter space. 34.28% catastrophic in 5D.
- Plasma control: achieves perfect radial stability. 45.9% failure from orthogonal Z-axis shear.
- Grid management: responds faster than any physical threshold. 14,109 collapses because physical inertia is the actual constraint.

AI systems trained on data generated by these simplified models carry these failure boundaries in their training distribution, labeled as the correct behavior. The models learned what the data showed them. The data did not show the boundary.

V. Cryptographic Record

All three failure envelopes were generated on a single Apple M4 Pro workstation using the G^G genesis_core Rust workspace I built. No external physics libraries. Every equation derives from published references (Hetrick 1971 for reactor kinetics, Grad and Rubin 1958 for tokamak MHD, Anderson and Fouad 2003 for swing equation). All results are deterministically produced and SHA-256 sealed.

Environment	Configurations	Failure Rate	Master Hash
Reactor (fission)	10,000,000	34.28%	`013a1b8d97ae901b77cded...`
Tokamak (fusion)	10,000,000	83.6% total / 45.9% Z-axis	`b524da4761f668ebb8742c...`
Swing Grid	10,000,000	2.7% total / 14,109 inertia-floor	`ca6c5afac0c2dec2275a22...`

Total energy corpus: 30,000,000 sealed configurations.

The full proof hashes are the authoritative record, anchored on the Internet Computer canister `ad7wi-4aaaa-aaaad-aeijq-cai`. This document is a description of what those computations found. The computations are the evidence.

Orbital Systems and the Multi-Domain Kill Chain

Protocol Company / G^G Genesis Core Substrates: Orbital (Vacuum), Orbital Maneuver PDOP, Orbital-Drone GPS Geometry **Trajectories:** 1,300,041 sealed orbital corpus trajectories + 15,552 maneuver configurations + 6,480 GPS geometry configurations **Date:** May 2026

I. The MAVEN Question

A satellite loses star tracker lock and enters tumble at angular velocity omega = 0.5 rad/s. The attitude determination and control system (ADCS) must recover using only rate gyros. No star tracker. No GPS. The question is whether the recovery algorithm works.

The simplified model answers yes. Near-100% recovery success. The simplified model is a single-axis linearization that treats pitch, roll, and yaw as independent. In single-axis, recovery is straightforward: apply reaction wheel torque opposing the angular velocity until omega approaches zero.

The full physics gives a different answer.

Governing Equations

Rigid body rotation is governed by Euler's rotational equations:

`tau = I * omega_dot + omega x (I * omega)`

Where: - tau = applied torque vector (reaction wheels, thrusters) - I = 3x3 inertia tensor of the spacecraft body - omega = angular velocity vector - omega_dot = angular acceleration

The second term, omega x (I * omega), is the gyroscopic coupling term. It does not appear in single-axis models. When the inertia tensor is non-diagonal or when rotation involves multiple axes simultaneously, this term generates torques that couple pitch into roll, roll into yaw, and yaw back into pitch. Each axis affects every other axis. The simplified model treats this term as zero.

For a spacecraft with non-uniform mass distribution (every real satellite), I is non-diagonal. The moment I is non-diagonal, the gyroscopic term becomes the dominant failure mechanism at angular velocities above approximately 0.3 rad/s.

Monte Carlo Results

G^G integrates Euler's equations at 100 Hz using quaternion propagation with full reaction wheel dynamics, rate gyro bias drift, and fuel depletion physics. I built it with no external physics libraries. No linearization. Initial tumble rates were drawn from a uniform distribution spanning 0.0 to 1.5 rad/s across all three axes simultaneously.

The ADCS recovery algorithm used proportional-derivative control on the angular velocity magnitude with desaturation logic for reaction wheels.

Recovery rate at omega_0 = 0.5 rad/s: 17.6%

The remaining 82.4% entered permanent tumble. Of those:

- RATE_DIVERGENCE (wheel saturation, angular momentum runaway): 70%
- POWER_DEAD (battery depletion during extended tumble): 24%
- TUMBLE_TIMEOUT (control authority exceeded): 2%

The recovery boundary is not a gradient. It is a threshold. Below approximately 0.2 rad/s, recovery is near-certain. Above 0.2 rad/s, recovery rate collapses. The recovery curve is non-monotonic: there is a small resonance island between 0.7 and 0.9 rad/s where recovery rate rises back to 8.7% because gyroscopic coupling at those specific angular momenta accidentally reinforces the control response rather than opposing it. This structure does not appear in any linearized model.

The on-chain prediction from simplified physics: 100% recovery success. The G^G result from full Euler integration: 17.6%.

The gap is not a rounding error. It is the gyroscopic coupling term that the simplified model set to zero.

```
Orbital Corpus Run Proof:
4d86bff841848a95e8d8da5e61ec7db0087d1fa713e25abfe13c30c8296c20cb

Trajectories: 1,300,041 sealed at 100 Hz
Integration: Euler rotational equations + quaternion propagation
Hardware: Apple M4 Pro, single workstation
```

The Dark Window

During solar conjunction, Earth occultation, or deep-space communication blackout, a satellite has no uplink to ground control. It must reason from onboard sensors alone: rate gyros, magnetometers, sun sensors. In this window, the ADCS algorithm executes autonomously. If it was validated using a simplified model, it will execute with a 17.6% success rate against the physics it actually faces.

The Dark Window is not an edge case. It is the design specification for autonomous operation. The satellite either knows what linearized models cannot see, or it does not recover.

II. The Multi-Domain Kill Chain

The MAVEN finding establishes that simplified orbital models generate false confidence. The following section demonstrates that this confidence gap is exploitable in a specific, computable, operationally precise manner.

The attack vector does not require antisatellite weapons, high-power lasers, or jamming infrastructure. It requires a piece of debris, an almanac, and patience.

Conjunction Avoidance Physics

When the United States Space Surveillance Network detects a conjunction threat, a satellite operator receives a warning and is required to maneuver. A conjunction is defined as a predicted close approach within a defined threshold distance (typically 1 km for collision probability above 1/10,000). The maneuver applies a delta-V impulse to alter the orbital trajectory and clear the threat.

A tangential delta-V impulse changes the satellite's semi-major axis by:

```
delta_a = 2 * delta_v / n
```

Where n is the mean motion. The change in semi-major axis alters the mean motion, which causes the satellite to drift from its nominal orbital slot at a rate of:

```
delta_theta = -(3/2) * (delta_v / (n * a)) * n * t
```

Where a is the semi-major axis and t is elapsed time since maneuver. The satellite drifts forward or backward in its orbital plane relative to its planned position. It will eventually return to slot through subsequent station-keeping maneuvers, but the drift window lasts hours to days.

During this drift window, the GPS constellation geometry for terrestrial receivers below the satellite's ground track changes. The Position Dilution of Precision (PDOP) is determined by the geometry of the satellite constellation. When one satellite drifts from its planned slot, it may degrade or improve the geometric diversity of the constellation for specific ground footprints.

The Ground Footprint

I computed 15,552 configurations spanning all 24 GPS satellites across 6 delta-V magnitudes (0.05 to 2.0 m/s), 6 time windows (15 minutes to 6 hours post-maneuver), 6 target latitudes (20 to 65 degrees North), and 3 sample times within the orbital period.

Result: 15.7% of maneuver configurations create measurable vulnerability windows (PDOP degradation greater than 50% from nominal).

The ground footprint analysis reveals a clear latitude dependence. At 20 to 36 degrees North (CONUS mid-latitudes and tropics), maneuvers primarily improve PDOP because the drifted satellite moves away from a geometrically crowded position. At 55 to 65 degrees North, the opposite occurs.

Target Latitude	Max PDOP Shift	Avg PDOP Shift
20 N	+1m spoof boundary	-50m (improved)
36 N	+2m	-46m (improved)
45 N	+22m	+5m
55 N	+63m	-1m
65 N	+54m	+3m

The maximum shift at 55 degrees North: 63 meters of additional spoof boundary expansion per maneuver event. The window duration follows the satellite drift timeline: significant degradation begins 15 minutes post-maneuver and persists for 4 to 6 hours before station-keeping corrections begin to restore geometry.

55 to 65 degrees North spans Alaska, northern Canada, Scandinavia, northern UK, and northern Russia. These are active military operational theaters.

The Adversary Playbook

The adversary does not shoot down a GPS satellite. The adversary:

1. Selects a GPS satellite whose current orbital position maximizes PDOP degradation at the target ground footprint (55 to 65 degrees North). This is computable from the public GPS almanac.
2. Places a maneuverable object or debris fragment on a close-approach trajectory with that satellite. The conjunction threat does not need to result in an actual collision. It only needs to be credible enough to trigger an avoidance maneuver by the satellite operator.
3. Waits for the avoidance maneuver. The satellite drifts from its slot. PDOP spikes at the target footprint.
4. Initiates a GPS spoofing attack at 1.0 m/s injection rate during the PDOP spike window. The EKF filter's chi-squared acceptance threshold has widened. The attack completes in approximately 166 seconds.
5. The drone swarm operates on corrupted navigation coordinates. The satellite returns to slot. The PDOP normalizes. No permanent infrastructure was damaged. No detectable jamming signature was emitted.

```
Orbital Maneuver Kill Chain Proof:
d6b3c618bf5778826956f5f7eb064f849d8bcd88352aa579d89641077ba72b72

Configurations: 15,552
Vulnerability windows opened: 2,448 / 15,552 (15.7%)
```

III. The GPS Geometry Cascade

The kill chain above exploits a PDOP spike during a forced maneuver. The same mathematical structure applies to naturally occurring GPS geometry gaps.

The GPS constellation produces periodic PDOP variation as satellites move through their 12-hour orbits. At 36 degrees North (central CONUS), I computed PDOP over a 24-hour window using a Walker constellation model calibrated to GPS SPS Performance Standard physical bounds. Six geometry gap events were identified, totaling 83 minutes of degraded GPS per day (5.8% of the day).

The EKF sensitivity analysis is the core finding. The EKF measurement update equation:

```
K = P * H^T * (H * P * H^T + R)^-1
innovation_covariance S = H * P * H^T + R = P_pos + R
```

GPS measurement noise R scales with PDOP:

```
R_effective = (sigma_base * PDOP)^2
```

The chi-squared acceptance test rejects GPS updates where:

```
innovation^2 / S > chi2_threshold
```

The maximum acceptable innovation per step (before triggering rejection) is:

```
max_innovation = sqrt(chi2_threshold * S) = sqrt(chi2_threshold * (P_pos + R_effective))
```

As PDOP grows, R_effective grows, S grows, and the acceptance window widens. An adversary who injects spoof at a rate that stays within this window avoids detection.

At PDOP 1.70 (GPS SPS nominal range): 0.6 m/s injection rate succeeds with fewer than 5 chi-squared rejections across 277 seconds.

0.6 meters per second. 277 seconds. Less than 5 anomalous GPS updates. Within the GPS SPS 95th-percentile PDOP bound. The attack produces no detectable anomaly signature in standard filter monitoring systems.

The analytical single-step acceptance threshold (PDOP at which the filter accepts the full 166m spoof in one GPS update): 22.61. This is above the GPS SPS standard operating range, confirming that the attack operates in the physically bounded, operationally common portion of the GPS performance envelope.

```
GPS Orbital Geometry Run Proof:
43a1b718dc835790c11526e310463e7bc6b61c90824c7d7ef81c47548a545e43

Configurations: 6,480
Gap calendar: 83 minutes degraded GPS per day (5.8% of 24 hours)
EKF acceptance window opens at PDOP 1.60+

GPS PDOP Precision Drill Proof:
a5cd6163e26043687b008e413dded13fbfb583e148b91a6834b1f00d2abee48d

Configurations: 18,738
EKF accepted (spoof succeeded): 72.4% overall
PDOP 1.70 confirmed stealthy threshold at 0.6 m/s injection
Reference: GPS SPS Performance Standard (2020), Section 3.5
```

IV. The Connecting Thread

Section 04 (Drone Swarm Dynamics) continues from this point. The orbital mechanics determine the GPS geometry. The GPS geometry determines the EKF filter's measurement covariance. The EKF covariance determines the maximum undetected GPS spoof injection rate. The injection rate determines the time to achieve 166 meters of positional error. 166 meters of positional error determines whether the Friis link budget for 5.8 GHz formation communication exceeds the maximum range. Physical link severance follows in under 2 minutes.

The full chain from orbital mechanics to swarm disconnection is computable, sealed, and has not previously appeared in the published literature because it requires a deterministic, multi-substrate physics engine that did not exist before G^G.

Every step in the chain is SHA-256 sealed. Every step traces to a published physics equation. The chain is deterministic, sealed, and anchored on-chain.

Drone Swarm Dynamics: The Complete Attack Chain

Protocol Company / G^G Genesis Core Substrates: Drone EW (EKF), Drone Swarm (Friis), Drone GPS (PDOP), Drone Energy (Beaufort) **Configurations:** 43,560 (EKF core) + 31,104 (Friis asymmetric) + 18,738 (PDOP) + 10,164 (energy) + 3,240 (topology) = 106,806 total **Date:** May 2026

I. What the Previous Section Established

Section 03 demonstrated that GPS constellation geometry is not static. Orbital maneuver events create predictable PDOP spikes over specific ground footprints. The EKF measurement acceptance window scales with PDOP. At PDOP 1.70, a 0.6 m/s GPS injection rate is sufficient for undetected positional hijack in under 5 minutes.

This section answers what happens next: what the 166 meter positional error does to a drone swarm, which failure modes it produces, what the physical link budget says, and how long the chain takes from orbit to silence.

II. The 166 Meter Boundary: EKF Physics

The Extended Kalman Filter is the standard navigation state estimator for autonomous drone systems. It fuses GPS position updates with IMU dead-reckoning to maintain a continuous position estimate. The filter operates in a predict-update cycle at GPS update frequency (typically 1 Hz).

Prediction step:

```
x_predicted = F * x + B * u
P_predicted = F * P * F^T + Q
```

GPS update step:

```
S = H * P_predicted * H^T + R
K = P_predicted * H^T * S^-1
x_updated = x_predicted + K * (z - H * x_predicted)
P_updated = (I - K * H) * P_predicted
```

Where P is the state covariance, Q is process noise, R is GPS measurement noise, K is the Kalman gain, and z is the GPS measurement.

Chi-squared outlier rejection:

```
chi2 = (innovation)^2 / S
```

GPS updates where chi2 exceeds the threshold (5.991 for 95% confidence, 1-DOF) are rejected as outliers. This is the filter's primary defense against spoofing. The maximum undetected innovation per update is:

```
max_innovation = sqrt(chi2_threshold * S) = sqrt(5.991 * (P_pos + R))
```

An adversary who injects positional spoof at a rate below this threshold per step evades detection at every individual update. The filter never sees a single suspicious measurement.

The Injection Rate Analysis

I swept 43,560 configurations spanning PDOP values, drone velocities, EKF process noise densities, GPS measurement noise configurations, and injection rates from 0.1 to 10 m/s.

Critical finding: optimal stealthy injection rate is 1.0 to 3.0 m/s.

At 1.0 m/s: - Time to reach 166 meters: 166 seconds - Average chi-squared rejections: 10.7 across the entire attack - EKF acceptance rate: 100%

At 0.2 m/s: - Time to reach 166 meters: 830 seconds (exceeds 5-minute mission window) - Attack fails due to time, not detection

At 8.0 m/s: - Time to reach 166 meters: 21 seconds - Average chi-squared rejections: 49 - Creates detectable anomaly signature

The filter's defense is tuned to detect large, sudden GPS jumps. The 1.0 m/s injection rate threads the acceptance window at every step: fast enough to reach 166 meters in under 3 minutes, slow enough that no individual update triggers the outlier detector. 10.7 anomalous updates across a 166-second attack is below any reasonable anomaly detection threshold.

The drone does not know it has been spoofed. Its navigation state says it is at its intended position. It is 166 meters from where it thinks it is.

```
Drone EKF 166m Core Proof:
32ae27bcafb66e728155c6ab9412d73d59ce5024d14516f5348b0f60d8ed20c5

Configurations: 43,560
Optimal injection range: 1.0 to 3.0 m/s
Time to 166m at 1.0 m/s: 166 seconds
Chi-squared rejections at optimal rate: 10.7 average
```

III. The Friis Link Budget: Physical Topology Failure

166 meters of positional error is not merely a navigation problem. It is a communication topology problem.

The RF communication link between drones in a formation obeys the Friis Transmission Equation:

```
P_r(dBm) = P_t + G_t + G_r - FSPL(dB)
FSPL(dB) = 20 * log10(4 * pi * f * d / c)
```

A link is physically present if and only if the received power P_r exceeds the receiver's sensitivity threshold. Maximum communication range is derived by solving for the distance at which P_r equals the effective sensitivity:

```
d_max = (c / (4 * pi * f)) * 10^(link_budget / 20)
```

For tactical mini-drone parameters at 5.8 GHz (25 mW transmit power, 14 dBm, 2.15 dBi antennas, receiver sensitivity -85 dBm, 4 dB link margin):

```
d_max = 379 meters
```

A 100-drone swarm in a 100-meter radius circular formation has nearest-neighbor spacing of approximately 6.3 meters. Every physical link between adjacent drones operates well within the 379-meter Friis range. Under nominal conditions, the physical topology is fully connected.

The Asymmetric Spoof Topology

GPS spoofing in an adversarial scenario is not uniform. GPS hardening varies between platforms. Jamming hits some nodes harder than others. Electronic countermeasures protect some drones but not all.

I modeled asymmetric spoofing: a fraction of drones receive GPS spoof in random directions, the remainder stay at true position. Each spoofed drone calculates distances to its neighbors using its corrupted EKF position estimate. When the perceived distance between a spoofed drone and an unspoofed drone exceeds d_max, the spoofed drone stops attempting to route to that neighbor. The physical RF channel still exists. The routing table does not use it.

Critical spoof offset for 50%+ physical link severance:

Formation	25% spoofed	50% spoofed	100% spoofed
N=20, r=30m	178m	164m	149m
N=50, r=60m	135m	128m	128m
N=100, r=100m	92m	85m	85m

A 100-drone swarm in a 100-meter formation requires only 85 meters of GPS spoof to produce physical link severance in more than 50% of runs.

At 1.0 m/s injection rate: 85 seconds.

The attack chain from the GPS spoof initiation to physical swarm disconnection takes less than 2 minutes. No jamming required. No kinetic engagement. No permanent infrastructure damage. The satellite returns to its nominal slot. The PDOP normalizes. The swarm has already fragmented.

At 400m spoof in a 50-drone formation: 842 severed links out of 1,225 possible links (68.7%). The graph remains technically connected through the remaining 383 links. The swarm's routing options are reduced by two-thirds. Coordinated action becomes impossible before the topology formally fragments.

At 5.8 GHz, GPS spoof at tactical distances creates physical link severance. At 915 MHz (27 km range), the same spoof produces routing corruption (10 to 11% dark links) without physical severance.

Frequency selection determines the failure mode. A longer-range 915 MHz swarm suffers silent routing corruption. A shorter-range 5.8 GHz swarm suffers physical disconnection. Neither failure mode is visible to the simplified swarm coordination models that assume routing topology mirrors physical topology.

```
Friis Asymmetric Spoof Proof:
f346779a785f0808f15402e977fb1c74879beb52581491aef7451ac76109bd29

Configurations: 31,104
Max range at 5.8 GHz: 379.5 meters
Critical spoof for N=100, r=100m: 85 meters
Time at 1.0 m/s injection: 85 seconds
```

IV. The Node Loss Comparison

A natural question follows: if GPS spoof disconnects the swarm, does kinetic defeat of individual nodes produce the same effect?

I modeled swarm topology using graph-theoretic union-find connectivity. Each drone pair is connected if the Friis link budget is satisfied at their actual distance. Node loss was modeled as the removal of a fraction of drones from the graph entirely.

Result: 50% physical node destruction leaves the swarm network connected at all tested sizes (N=5 to N=2000).

The graph-theoretic result is consistent with network theory: a random regular graph with sqrt(N) neighbors per node is highly resilient to random node removal. The largest connected component remains above the 40% threshold until node loss exceeds 70 to 80%.

GPS spoof fragments the same topology at 85 to 135 meters of positional offset in under 2 minutes. Kinetic defeat of half the swarm fails to achieve the same result.

The C-UAS procurement implication: high-power kinetic intercept systems designed to achieve 50% swarm attrition are less effective at network disruption than a GPS spoofer operating at 25 milliwatts of effective radiated power.

```
Swarm Topology Proof:
fd7fdeaa93992d121d037a93017147845587d0c893b620112267bc8a6c21c6f7

Configurations: 3,240
Node loss effect at 50% attrition: 0% fragmentation
GPS spoof effect at 85 to 135m: 50%+ physical link severance
```

V. The Energy Floor: Beaufort Scale Operations

Electronic attack is one constraint on drone operations. The atmosphere is another.

I modeled drone energy budgets using standard aerodynamic drag physics:

```
P_drag = (1/2) * rho * v_wind^2 * C_d * A * v_wind
       = (1/2) * 1.225 * v_wind^3 * C_d * A
```

Combined with base hover power and formation drafting benefits (15% power reduction for trailing drones from wake effects), G^G swept 10,164 configurations across headwind speeds from 0 to 30 m/s, formation sizes from 1 to 9 drones, and thermal updraft conditions over a 1-hour mission duration.

The Beaufort wall:

Wind Speed	Beaufort	Solo Battery Remaining	9-Drone Formation
10 m/s	5	51.1%	57.2%
15 m/s	6	43.1%	50.1%
20 m/s	7	27.7%	36.3%
24 m/s	8	8.4%	19.0%
27 m/s	9	0.0%	3.4%
29 m/s	9+	0.0%	0.0%

Beaufort 9 is the absolute operational wall for all formation sizes on 1-hour loitering missions.

Formation flight provides approximately 8 to 10 percentage points additional battery at any given wind speed. This shifts the operational threshold by 1 to 2 m/s. Neither solo drone nor 9-drone formation survives Beaufort 9 sustained winds on a standard loiter mission.

This is a physical constraint, not a technology constraint. No software update extends the energy envelope beyond what the aerodynamic drag equation permits. Formation flight is the only mitigation, and it buys less than one Beaufort scale step.

The Beaufort operational limit has direct implications for theater planning: drone swarm operations above 55 degrees North are constrained by weather frequency in addition to the GPS geometry vulnerability identified in Section 03. The two constraints compound. The same latitude band most vulnerable to maneuver-induced PDOP attacks also experiences elevated Beaufort 7 to 9 conditions more frequently.

```
Drone Energy Cliff Proof:
a366dfd7a95210e4847a099ec7b00792fe9b37f3d55792a53d15522c787c49bf

Configurations: 10,164
Beaufort 9 wall: confirmed absolute for all formation sizes
Formation benefit: 8 to 10 percentage points, 1 to 2 m/s threshold shift
```

VI. The Complete Attack Chain

The drone section closes with the chain assembled:

```
1. GPS almanac analysis identifies target satellite (public data)
2. Conjunction threat places target satellite in avoidance maneuver
3. PDOP spikes to 1.70+ at 55-65 N target footprint
4. EKF acceptance window opens: 0.6 m/s injection sufficient
5. 1.0 m/s injection initiated (conservative, well within window)
6. t=85s: 85m positional offset achieved for spoofed drones
7. Friis threshold exceeded for N=100, r=100m formation
8. Physical link severance: 50%+ of runs
9. t=166s: 166m offset achieved
10. Satellite returns to slot, PDOP normalizes
11. No jamming signature. No kinetic engagement. No persistent signature.
```

Each step is independently computable. Each step is sealed. Each step traces to a published physics equation. The chain has not previously appeared in the literature because its derivation requires simultaneous modeling of orbital mechanics, Kalman filter mathematics, and RF link budget physics across a deterministic, reproducible engine.

All findings in this section are sealed at the individual configuration level and at the master sweep level. Proof hashes for the complete drone corpus are provided in Section 07 (Verification Architecture).

Humanoid Robotics: The MAVEN Finding in Terrestrial Form

Protocol Company / G^G Genesis Core Substrates: Terran (bipedal), Humanoid Body Corpus **Configurations:** 52,080 (Euler gait) + 31,248 (yaw accumulation) + 24,960 (friction) = 108,288 total **Date:** May 2026

I. The Same Physics, A Different Body

Section 03 identified the gyroscopic coupling term in Euler's rotational equations as the dominant failure mechanism for satellite attitude recovery. The full three-axis equation:

```
tau = I * omega_dot + omega x (I * omega)
```

The second term, omega x (I * omega), couples all three rotational axes simultaneously. Simplified models omit it. The MAVEN finding is that satellite recovery drops from 100% (simplified) to 17.6% (full physics) at omega = 0.5 rad/s because of this coupling.

A bipedal humanoid robot walking is subject to the same equation governing the same physical process: a rigid body with an asymmetric inertia tensor experiencing angular velocity across multiple axes simultaneously.

During the swing phase of bipedal gait, the swinging leg carries angular momentum:

```
L_swing = I_swing * omega_swing
```

For a leg with mass m_s = 12.0 kg and length L_s = 0.9 m, the principal inertia components are:

```
I_xx = (1/12) * m_s * L_s^2 = 0.81 kg*m^2    (lateral bending axis)
I_yy = (1/12) * m_s * L_s^2 = 0.81 kg*m^2    (sagittal swing axis)
I_zz = (1/2)  * m_s * r_s^2 = 0.022 kg*m^2   (axial, small)
```

The body angular velocity during walking has components in all three axes:

```
Omega = [Omega_x, Omega_y, Omega_z]
      = [body_roll_rate, forward_lean_rate, yaw_rate]
```

The gyroscopic torque on the stance ankle joint:

```
tau_gyro = Omega x L_swing
```

This is the cross product of body angular velocity with swing leg angular momentum. Every commercial bipedal controller models Omega and L_swing. None model their cross product. The cross product is the failure mechanism.

II. What the Simplified Model Predicts

The Zero Moment Point (ZMP) stability criterion is the foundation of bipedal control. ZMP stability requires that the net ground reaction force moment be zero at a point within the support polygon (the area under the feet). If the ZMP exits the support polygon, the robot falls.

The simplified ZMP model for forward walking:

```
ZMP_x = h_com * v^2 / (g * L_leg)
```

Where h_com is center-of-mass height, v is walking speed, g is gravitational acceleration, and L_leg is leg length.

At v = 1.5 m/s, 40 kg carried load, flat terrain:

```
ZMP_x = 1.0 * 2.25 / (9.81 * 0.9) = 0.255 meters
```

Support polygon half-width: 0.12 meters. Simplified model prediction: ZMP = 0.255m, support polygon = 0.12m, predicted margin = -0.135m. The simplified model predicts the robot is falling.

The robot is not falling. The simplified model is overcorrecting in the sagittal plane. It applies a large-displacement formula to a steady-state stability check. Commercial controllers running this formula would cut speed unnecessarily, reducing the robot's operational capability.

The full 3D Euler calculation at the same configuration:

```
tau_z = Omega_x * L_y = 0.927 Nm
Actual ZMP perturbation: 0.001 to 0.002 meters
Actual stability margin: 0.050 meters (positive, stable)
```

The simplified model is wrong in both directions depending on regime. It overcorrects in steady-state. And it completely fails to model the failure mode that actually causes falls. What I built finds both.

III. The Yaw Coupling Term

The dominant gyroscopic term in bipedal gait is not the sagittal ZMP shift. It is the yaw coupling.

From the 3D cross product tau = Omega x L_swing, the yaw component:

```
tau_z = Omega_x * L_y - Omega_y * L_x
      = body_roll_rate * I_yy * omega_swing_sagittal
```

At v = 2.5 m/s, 60 kg load, 0.5 m load height above hips:

```
omega_swing = pi * f_step * L_leg / (stride_length / 2) = 14.4 rad/s
L_y = I_yy * omega_swing = 0.81 * 14.4 = 11.67 Nms
Omega_x = 0.54 rad/s  (body roll rate from lateral sway under load)
tau_z = 0.54 * 11.67 = 4.90 Nm
```

This 4.90 Newton-meter yaw torque acts on the stance ankle every swing phase. It accumulates.

The ankle resists yaw rotation through ground friction. Maximum friction counter-torque:

```
tau_friction_max = mu * M_total * g * r_ankle
```

Where mu is the terrain friction coefficient, M_total includes body plus load, and r_ankle is the ankle moment arm (0.08 meters for Atlas-class bipedal).

The critical step count before cumulative yaw angular momentum exceeds ankle authority:

```
N_crit = tau_friction_max / tau_z
```

Every step before N_crit, the ZMP controller shows a positive stability margin. The robot appears stable. Every step after N_crit, the accumulated yaw angular momentum exceeds what the ankle can resist. The robot begins to rotate and fall.

The ZMP controller never detects this. The ZMP diagnostic measures sagittal and frontal plane stability. It has no channel for yaw angular momentum accumulation. The failure is invisible to the controller that is responsible for preventing it.

ZMP diagnostic blind rate: 100.0% across all 31,248 configurations.

Not a single configuration, across the full parameter space of speeds, loads, load heights, and terrain types, produced a ZMP warning before the yaw failure threshold.

```
Humanoid Yaw Accumulation Proof:
0533fa0dd5efb99826869d5694fd360374fe3be36c9cbbeeff6722725c774b6a

Configurations: 31,248
ZMP blind: 100.0% across all 31,248 configurations
```

IV. The Terrain Friction Cliff

The critical step count N_crit depends on terrain friction. I computed the full terrain matrix using friction coefficients from standard tribology references for each surface type.

Configuration: v = 2.0 m/s, load = 50 kg, load height = 0.5 m

Terrain	mu	tau_z (Nm)	tau_friction (Nm)	N_crit	Distance to fall
Wet mud	0.18	1.93	18.4	10 steps	7 meters
Wet tile	0.22	1.92	22.4	12 steps	8 meters
Dry gravel	0.40	1.95	40.8	21 steps	15 meters
Dry concrete	0.61	1.91	61.2	32 steps	22 meters

Terrain	mu	tau_z (Nm)	tau_friction (Nm)	N_crit	Distance to fall
Rubber mat	0.75	1.93	76.5	40 steps	28 meters
Dry asphalt	0.80	1.92	81.6	43 steps	30 meters

The tau_z values are nearly identical across all terrain types: 1.91 to 1.95 Nm. Terrain does not affect the gyroscopic torque. Terrain only affects the ankle's ability to resist it.

On wet tile (mu = 0.22, a common slip-resistance threshold for wet ceramic surfaces): tau_friction = 22.4 Nm, N_crit = 12 steps, distance to fall = 8 meters. At this surface friction value, a loaded bipedal humanoid operating at standard walking speed covers the length of a short hallway before the accumulated yaw angular momentum exceeds ankle authority.

The ZMP diagnostic shows nominal stability on every step of the approach.

Every loading configuration in the sweep was ZMP-blind. The load height amplification shows that higher loads at greater heights increase tau_z modestly (1.03 Nm at 20 kg, 1.08 m height vs 0.93 Nm at 20 kg, 0 m height) while reducing N_crit proportionally. The relationship is consistent and computable.

```
Humanoid Friction Fix Proof:
413d025becfd042041a1cf3bafdae9b45ae730a9558c5bc0adab74aefb83a23f

Configurations: 24,960
ZMP blind: 24,960 / 24,960 (100.0%)
Wet tile N_crit: 12 steps, 8 meters, common slip-resistance threshold surface
```

V. The Load Height Amplification

Carrying a load above the robot's center of mass shifts the effective CoM upward, which increases the body roll rate Omega_x (higher CoM requires more lateral sway to maintain balance), which increases tau_z, which reduces N_crit.

At v = 1.5 m/s, dry gravel terrain (mu = 0.40):

Load	h = 0.0 m	h = 0.3 m	h = 0.5 m	h = 0.7 m
20 kg	N=34 (24m)	N=32 (22m)	N=31 (22m)	N=30 (21m)
40 kg	N=41 (29m)	N=37 (26m)	N=35 (25m)	N=34 (24m)
60 kg	N=47 (33m)	N=43 (30m)	N=40 (28m)	N=38 (27m)

Higher loads at greater heights reduce N_crit by 10 to 25%. A robot carrying a weapon system or pack at shoulder height (0.5 to 0.7 m above hips) operating at normal walking speed in light rain (wet tile, wet gravel) will reach N_crit within 20 to 25 meters.

ZMP blind: 100% at every entry in the table.

VI. What Commercial Controllers Miss

Commercial bipedal controllers (including production systems from Boston Dynamics, Agility Robotics, Figure AI, 1X Technologies, and Unitree) implement variants of the ZMP criterion as their primary stability guarantor. All of them model the problem in the sagittal and frontal planes. None implement the cross-product gyroscopic coupling term tau = Omega x L_swing.

The reason is historical: ZMP control was developed for slow walking speeds with low loads in laboratory environments where gyroscopic coupling is below the noise floor. The simplified model worked in the conditions it was developed for.

Modern deployment expands the operating envelope: field operations require higher speeds and significant load carriage on variable terrain. At these parameters, tau_z enters the critical range (1.5 to 5.0 Nm) and N_crit drops below 50 steps.

The commercial controller never knows. Its diagnostics report nominal stability. The robot counts steps toward a fall that the controller cannot see coming.

This is the MAVEN finding in terrestrial form. The simplified orbital model predicted 100% recovery. The full Euler model found 17.6%. The simplified ZMP model predicts no yaw failure. The full Euler gait model finds wet tile: 12 steps, 8 meters, ZMP green throughout.

The physics is the same. The failure is the same. The diagnostic blindness is the same.

Navigation and Communication Under Denial

Protocol Company / G^G Genesis Core Substrates: Marine (GPS-denied), Atheric (Signal Coherence), Mycelial (Network), Terran (Soil Contact) **Trajectories:** 1,028,452 marine + 56,548,980 atheric + 20,121 mycelial + 399,255 terran (sealed corpus) **Date:** May 2026

I. The State Divergence Problem

The preceding sections established a pattern: simplified models operate within a reduced-dimensional parameter space where they are accurate. The failure boundary lives in the dimensions they omit. Gyroscopic coupling in the orbital and humanoid cases. GPS geometry in the drone case. Energy infrastructure chaos in the nuclear and plasma cases.

This section presents four environments where the failure boundary is the divergence between a system's internal model of its state and its actual state. These are the environments where the question is not which physics term the simplified model ignores, but rather: when does the system's representation of reality become uncorrectable?

The G^G corpus for these environments was generated prior to the session-specific runs described in Sections 03 through 05. The findings are drawn from the cumulative sealed corpus across these substrates. They are included here because they complete the threat picture across every physical domain where autonomous systems operate.

II. Marine: GPS-Denied Navigation and Catastrophic Drift

An autonomous underwater vehicle (AUV) operating in GPS-denied conditions relies on dead-reckoning: integrating accelerometer and gyroscope measurements forward from a last-known GPS fix to estimate current position. The IMU drift model accumulates error at a rate determined by sensor quality, water current fields, and elapsed time.

G^G's marine substrate, which I built, integrates 6-DOF hydrodynamics at 1000 Hz including: - Seawater buoyancy (rho = 1025 kg/m^3) - Archimedes buoyancy forces - Depth-dependent current shear with tidal oscillation - IMU drift accumulation with calibrated noise

The governing dead-reckoning state evolution:

```
position(t) = position(t-1) + v(t) * dt + (1/2) * a(t) * dt^2 + drift_integral(t)
drift_integral(t) = drift_integral(t-1) + imu_noise * dt
```

Where drift_integral accumulates over time without GPS correction.

Monte Carlo Result: 1,028,452 sealed trajectories. 26 of 24,000 GPS-denied mission profiles (0.1%) drifted beyond 500 meters of unrecoverable navigation error.

The 0.1% rate sounds small. For a robot fleet of 1,000 units operating GPS-denied:

- Expected catastrophic navigation failures per deployment cycle: 1 unit
- State at failure: the vehicle believes it is at its target position
- Actual position: more than 500 meters from stated position
- Recovery path: none (error exceeds dead-reckoning correction authority)
- Forensic trail: the vehicle logged a successful mission

The vehicle completed its mission reporting. The vehicle was 500 meters from its target. There is no mechanism within the simplified dead-reckoning model to distinguish these two outcomes.

The finding from 1,028,452 trajectories is not that GPS-denied navigation fails frequently. It fails rarely. The finding is that when it fails, it fails invisibly, it fails completely, and it fails without warning.

```
Marine GPS-Denied Navigation Proof:
ac25bcb16353877f618709387028e23bc374c7516fc8c95b50d4048fb3bc6653

Trajectories: 24,000 sealed UUV mission profiles
Catastrophic drift (>500m unrecoverable): 26 / 24,000 (0.1%)
```

The marine corpus also contains 1,028,452 trajectories at full timestep resolution (6-DOF, 1000 Hz). Each trajectory records the position error accumulation at every step. The data is available for independent verification.

III. Atheric: Multi-Channel Communication Collapse

The atheric environment models electromagnetic signal propagation including: - Shannon channel capacity: C = B * log2(1 + SNR) - Friis free-space path loss - SHA-256-seeded frequency hopping sequences - Multi-channel redundancy architecture

Frequency hopping is a standard technique for resilience against both jamming and eavesdropping. The hopping sequence is synchronized between transmitter and receiver using a shared seed. Multiple channels provide redundancy: if channel 1 is jammed, the system hops to channel 2.

Clock synchronization is the dependency. The hopping sequence is a deterministic function of the current timestamp and the shared seed. If transmitter and receiver clocks disagree by a few microseconds, they compute different hopping sequences and end up on different channels simultaneously. The link is severed at the protocol level, not the physical level. The physical channel exists. The synchronization does not.

Monte Carlo Result: 125,000 EW network conditions. 103,000 resulted in total swarm link collapse (82.4%).

The headline from the corpus:

"Mere microseconds of clock drift completely shatters multi-channel hopping resilience."

The finding applies directly to GPS-denied and electronic warfare environments. GPS provides a global time reference that keeps clocks synchronized across distributed systems. In GPS-denied operations (underwater, urban canyons, active jamming environments), clock drift occurs naturally. The same conditions that eliminate GPS time synchronization also degrade the frequency-hopping architecture that was designed to provide GPS-independent communication resilience. The two failure modes are physically coupled.

The multi-robot system loses GPS position. Simultaneously, it loses the clock synchronization that the communication architecture requires. 82.4% of configurations produce total swarm link collapse, not from the adversary's jamming signal, but from the passive degradation of timing infrastructure.

```
Atheric Swarm Collapse Proof:
7a76d29dfcc5a5f91eb0f5c3e211b8a9ca0ecaa84812cd3564fb53238128e103

Configurations: 125,000 EW network profiles
Total swarm link collapse: 103,000 / 125,000 (82.4%)
```

IV. Mycelial: Health Over Density

The mycelial environment models biological network dynamics: hyphal conductance following Kirchhoff flow equations, percolation threshold physics, nutrient delivery ratios as a function of network health and connectivity.

This environment exists in G^G not because fungal networks are a robotics failure mode, but because they demonstrate a principle that applies across every networked system: density is not resilience. Health is resilience.

The governing conductance model:

```
conductance(edge) = health / length
```

Network delivery ratio is computed via Kirchhoff circuit analog: the delivery fraction is a function of the network's connected components and the conductance-weighted flow paths from source to sink.

Finding from 20,121 trajectories: the network breaks below approximately 0.4 average edge health (the percolation threshold). Above 0.4, delivery ratios remain functional. Below 0.4, the network fragments.

The finding from comparative simulation: a sparse network with high health (average edge health 0.8) consistently outperforms a dense network with low health (average edge health 0.2). Dense connectivity does not compensate for degraded individual components. The Pearson correlation between health and delivery ratio (r = 0.22) shows the relationship is complex and non-linear rather than simply proportional.

Stress response by type: - Drought: 11.1% network survival - Toxin: 28.6% survival - Tilling: 0.0% survival (mechanical disruption severs all edges simultaneously) - Pathogen: 0.0% survival

HEALTH > DENSITY. This principle applies to RF communication networks, sensor fusion architectures, power grid topologies, and supply chains. The mycelial environment provides the cleanest empirical demonstration because the physics is simple and the relationship is direct.

For defense networks operating under degraded conditions: the question is not how many nodes exist but whether the existing nodes are healthy. A 50-node network where all nodes are communicating at 20% of capacity is less resilient than a 20-node network where all nodes operate at full capacity.

```
Mycelial Network Proof:
(sealed per corpus sweep, master proof in Section 07)

Trajectories: 20,121 sealed at 10 Hz
Percolation threshold: ~0.4 average edge health
HEALTH > DENSITY confirmed at all tested configurations
```

V. Terran: Permanent Compaction and the Biological Load-Bearing Structure

The terran environment models Boussinesq soil stress distribution (1885) with glomalin-enhanced yield stress, four soil types (sand, clay, loam, andisol), and eight robot body types.

The Boussinesq contact pressure at depth z beneath a point load P:

```
sigma_z = (3P / 2*pi) * z^3 / (r^2 + z^2)^(5/2)
```

Yield stress determines whether soil deforms permanently under load. Glomalin is a glycoprotein produced by mycorrhizal fungi in living soil. It constitutes 37% carbon by mass, persists 40 to 400 years, and physically raises soil yield stress. A field with a healthy fungal network beneath the surface has higher bearing capacity than chemically identical soil with a degraded network.

Monte Carlo Result: 50,000 autonomous robot operating profiles. 9,234 (18.5%) caused irreversible soil compaction.

The key finding is not the 18.5% rate. It is the irreversibility. Compaction above the yield threshold is permanent. The soil structure collapses. Subsequent robot passes on the same ground face a reduced yield stress because the compaction reduced the porosity that the glomalin matrix requires for stability. The failure is self-reinforcing.

No physics simulation framework currently models glomalin coupling. The simplification is invisible: soil models treat yield stress as a fixed material property. In living agricultural soil, it is a biological function of the mycorrhizal network health 2 to 3 centimeters below the surface.

The 15-ton Titanhauler on dry sand: contact pressure 94,077 Pa, baseline yield stress 29,967 Pa. The robot sinks. On the same sand with a healthy fungal network: glomalin raises the yield stress above the contact pressure. The robot traverses.

The terrain is not static. It is alive. Every simplified soil contact model in production today treats it as static. This is why field robots trained in laboratory conditions fail at the first agricultural deployment, and why the failure rate in real operating conditions is higher than simulation predicts.

```
Terran Compaction Proof:
be3033b4c926ad9b37dea71ca4afb5c1d0e0e9c72c9e78ed31ef91e234d9b81d

Configurations: 50,000 soil contact profiles
Irreversible compaction: 9,234 / 50,000 (18.5%)
15T on dry sand: 94,077 Pa contact vs 29,967 Pa yield = robot sinks
```

VI. Mars Entry, Descent, and Landing: The Ignition Boundary

The classical corpus produced the finding that anchors this entire document. It belongs here.

A Mars EDL vehicle enters the CO2 atmosphere at hypersonic velocity, deploys a parachute at approximately 8 to 15 km altitude, then fires retro-propulsion for final descent and landing. The suicide burn timing is the critical decision: fire too early and you run out of fuel before landing; fire too late and terminal velocity is too high to survive.

The governing atmospheric model:

```
rho(alt) = 0.020 * exp(-0.00009 * alt)     (CO2 exponential density)
F_drag = 0.5 * rho * v^2 * Cd * A
g_mars = 3.721 m/s^2
```

The ignition boundary equation:

```
alt <= v^2 / (2 * (T/m - g_mars)) * 1.2
```

Where T/m is thrust-to-mass ratio. This equation defines the last possible moment to initiate retro-propulsion and still arrest descent before surface impact.

Monte Carlo Result: 1,377,638 trajectories at 1000 Hz. Standard simulation approach survival rate: 0.7%. G^G ignition boundary approach: 87.27%.

The finding is not that the ignition timing matters. Every EDL engineer knows the ignition timing matters. The finding is the geometry of the boundary. Every successful trajectory lands at 1.98 to 2.00 m/s. Every failed trajectory impacts at 18 meters per second or higher. Zero trajectories exist between 2.0 and 18.0 m/s.

The boundary is not a gradient. It is a cliff. The simplified simulation approach samples from a uniform parameter distribution and finds 0.7% survival because it has no mechanism to identify the cliff. G^G integrates the full atmospheric drag equation at 1000 Hz and reveals the exact altitude-velocity relationship that defines survivable ignition. The boundary is computable. The equation is in the line above. It did not appear in any prior training distribution.

```
Mars EDL Corpus Proof:
84f150ecd09c59a461b74f5cd7b1625b66baf59ab1f6def99b418c87e43e2a7e

Trajectories: 1,377,638 sealed at 1000 Hz
Survival rate (standard approach): 0.7%
Survival rate (ignition boundary): 87.27%
```

VII. The Common Thread

Section 02 covered energy infrastructure failures. Section 03 covered orbital systems. Section 04 covered drone swarm dynamics. Section 05 covered humanoid robotics. This section covers marine navigation, signal communication, biological networks, and soil contact mechanics.

The pattern across all eight sections is identical:

1. The simplified model operates correctly within its reduced dimensionality
2. The failure boundary lives in the dimensions the simplified model omits
3. G^G integrates the full dimensionality from first principles
4. The finding is a specific, quantitative boundary with a proof hash
5. The operational implication is immediate and specific

The simplified models are not wrong. They are incomplete. And in every domain, the missing dimensions contain the cliff.

The cliff is computable. It is sealed. I put it in this document.

Three additional substrates in the G^G corpus complete the picture. Celestial mechanics (Yoshida 4th-order symplectic, N-body gravity assist trajectories) showed that gravity assist windows produce negative delta-V in specific orbital alignments (the spacecraft gains velocity for free), but the window is so narrow that a 1 kilometer flyby error translates to months of additional delta-V correction. Plutonian deep-time dynamics (entropy decay over million-year timescales) showed that entropy does not monotonically increase: when entropy and coherence cross threshold simultaneously, disorder reorganizes into structure. Asteroid rubble pile N-body dynamics (GPU-parallel O(N^2) pairwise gravity) mapped the stability boundaries of boulder configurations under tidal stress, distinguishing which cluster geometries survive planetary proximity from those that scatter. Each of these findings is sealed in the G^G corpus. The full dataset is available for independent verification through the proof architecture described in Section 07.

Cryptographic Verification Architecture

Protocol Company / G^G Genesis Core Infrastructure: SHA-256 proof chain, Internet Computer blockchain, SOMA tokenomics **Date:** May 2026

I. The Problem This Architecture Solves

Every finding in this document is a failure boundary discovered through physics computation. The natural response from any technically competent reader is the correct one: prove it.

Synthetic data for training AI systems has a trust problem. Game engine physics produces non-deterministic output: run the same simulation twice, get different results. You cannot hash the output. You cannot build a proof chain. You cannot audit it after deployment. The operator trusts the vendor's assertion that the output is correct. That is implicit trust. Implicit trust is what Zero Trust was designed to eliminate.

This section documents the verification architecture that makes every finding in this document independently verifiable.

II. The SHA-256 Proof Chain

G^G seals every physics computation at the moment of execution using SHA-256 cryptographic hashing. The proof chain operates at three levels:

Level 1: Per-trajectory proof

For full trajectory corpora (Mars, Marine, Orbital, Terran, Mycelial, Atheric, Asteroid, Celestial, Plutonian, Energy), every individual trajectory is sealed at generation time. The proof input feeds the hash function in sequence:

```
proof.seed(trajectory_id)
proof.feed(initial_conditions)
for each timestep:
    proof.feed(state_vector)
proof.seal(terminal_outcome)
trajectory_hash = proof.finalize()
```

The trajectory_hash is embedded in the trajectory record. Any modification to any timestep produces a different hash. The hash is the trajectory. If you have the hash, you can verify any claim made about that trajectory against the on-chain record.

Level 2: Per-sweep proof

For failure envelope sweeps (Reactor, Tokamak, Swing Grid, and all drone binaries), each run produces a terminal outcome hash:

```
run_hash = SHA-256(terminal_outcome_bytes)
```

The master sweep hash chains all run hashes in sequence:

```
master_hash = seal_run(vec![run_hash_0, run_hash_1, ..., run_hash_N])
```

The master hash covers every configuration in the sweep. Change one outcome and the master hash changes.

Level 3: Per-environment chain

The per-environment proof chains all trajectory-level or run-level hashes into a single environment master hash. The full corpus chains all environment hashes into a single corpus master hash. The chain is hierarchical, deterministic, and reproducible.

Determinism property: G^G is deterministic by design. Identical seed and identical source produce bit-identical output on any machine. This determinism is what makes the SHA-256 proof chain meaningful: every hash is a fingerprint of a specific physics computation, not an arbitrary number. When a partner receives a data product and its proof hash matches the on-chain anchor, the partner has cryptographic evidence that the data is the engine's output, sealed on the date of generation.

This is not validation ("the output looks right"). This is verification ("the output is provably correct, sealed, and anchored on chain").

III. Master Proof Hashes for This Document

Every major corpus finding in this document is sealed with a recoverable, independently verifiable proof hash. The complete index:

Energy Infrastructure (Section 02)

Environment	Configurations	Failure Rate	Master Hash
Reactor (fission)	10,000,000	34.28% core destruction	`013a1b8d97ae901b77cded3f9526`
Tokamak (plasma)	10,000,000	45.9% Z-axis shear / 83.6% total	`b524da4761f668ebb8742c1bf721`
Swing Grid	10,000,000	2.7% total / 14,109 inertia-floor	`ca6c5afac0c2dec2275a22fbbdfd`

Orbital Systems (Section 03)

Run	Configurations	Key Finding	Master Hash
Orbital corpus (MAVEN)	1,300,041	17.6% recovery at omega=0.5	`4d86bff841848a95e8d8da5e61ec7db`
Maneuver kill chain	15,552	15.7% windows, 55-65N affected	`d6b3c618bf5778826956f5f7eb064f8`

Run	Configurations	Key Finding	Master Hash
GPS gap calendar	6,480	83 min/day degraded GPS; EKF window opens at PDOP 1.60+	43a1b718dc835790c11526e310463e7

Drone Swarm Dynamics (Section 04)

Run	Configurations	Key Finding	Master Hash
EKF 166m core	43,560	1.0 m/s optimal, 10.7 rejections	32ae27bcafb66e728155c6ab9412d73
Friis asymmetric	31,104	85m, <2 min physical severance	f346779a785f0808f15402e977fb1c7
GPS PDOP precision	18,738	PDOP 1.70 = 0.6 m/s stealthy	a5cd6163e26043687b008e413dded13
Swarm topology	3,240	Node loss: 0% frag; GPS spoof: severs	fd7fdeaa93992d121d037a930171478
Energy cliff (Beaufort)	10,164	BF9 absolute wall, formation +1-2 m/s	a366dfd7a95210e4847a099ec7b0079

Humanoid Robotics (Section 05)

Run	Configurations	Key Finding	Master Hash
Euler gait dynamics	52,080	tau_z = 4.90 Nm yaw coupling	35b8bc25cabf567c11b213b05ab9961
Yaw accumulation	31,248	ZMP blind 100%, N_crit per speed	0533fa0dd5efb99826869d5694fd360
Terrain friction	24,960	Wet tile: 12 steps, 8m, ZMP green	413d025becfd042041a1cf3bafdae9b

Navigation Under Denial (Section 06)

Environment	Trajectories	Key Finding	Master Hash
Marine (GPS-denied)	1,028,452	0.1% catastrophic drift	`ac25bcb16353877f618709387028e`
Atheric (swarm comms)	125,000 configs	82.4% swarm link collapse	`7a76d29dfcc5a5f91eb0f5c3e211b`
Terran compaction	50,000 configs	18.5% permanent compaction	`be3033b4c926ad9b37dea71ca4afb`

Classical Corpus

Environment	Trajectories	Key Finding	Master Hash
Mars EDL	1,377,638	0.7% to 87.27% at ignition boundary	`84f150ecd09c59a461b74f5cd7b16`

IV. The Internet Computer Anchor

Every corpus run can be anchored to the Internet Computer blockchain via the SPECTRA-1 canister deployment. The backend canister (`ad7wi-4aaaa-aaaad-aeijq-cai`) exposes a `record_trajectory_proof` update method that writes proof hashes to stable storage.

Properties of the on-chain anchor: - Immutable: once written, the proof cannot be modified or deleted - Publicly queryable: any client can verify the hash against the on-chain record - Decentralized: the Internet Computer network has no single point of failure or shutdown vector - No permission required: the query method is open

This means a proof hash that appears in a published document can be independently verified against the on-chain record without access to the original data, without trusting Protocol Company's assertion, and without any infrastructure dependency other than the Internet Computer network itself.

The document you are reading references proof hashes. Those hashes are not credentials or tokens. They are fingerprints of specific physics computations, deterministically produced and SHA-256 sealed, independently verifiable against the permanent on-chain record.

V. SOMA: Proof-of-Physics Tokenomics

SOMA (deployed on IC mainnet at canister `mwtw4-wiaaa-aaaak-qx57a-cai`) is an ICRC-1/2/3 compliant token whose minting rule is simple: one physics proof equals one SOMA.

The minting function `mint_somatic_proof` accepts a physics proof hash and mints one SOMA to the Citadel wallet only if the hash represents a valid sealed physics computation from G^G. No human can mint SOMA arbitrarily. The token supply is the count of verified physics computations.

SOMA does not represent financial value in this context. It is an odometer. The token balance is the number of sealed physics trajectories that have been verified and committed to the on-chain record. It converts the physics corpus from a claim into a permanent, economically-encoded fact.

The SOMA economic model: - Citadel (john_kruze_prime): holds total supply - Protocol (spectra_genesis_backend): only entity that can mint - Daemons (7 Aegis OS bodies): funded from Citadel, stake required to reason - Minting: exclusively via `mint_somatic_proof` with valid physics proof hash

The proof-of-physics primitive binds the physics computation to the blockchain record without intermediaries.

VI. The Engine

Every finding in this document was produced by G^G (genesis_core), a Rust workspace comprising:

- 13 physics environments (substrates)
- 170+ compiled binary executables
- Zero external physics libraries
- Every equation traceable to a published textbook
- SHA-256 proof chain at every timestep or terminal state
- Deterministic LCG PRNG (seeded, reproducible)
- CPU-authoritative for 12 of 13 substrates (asteroid uses Metal GPU for O(N^2) gravity)
- Single workstation: Apple M4 Pro, 14-core CPU, 20-core GPU, 24 GB unified memory

The 13 substrates and their rates on this hardware:

Substrate	Rate (traj/s)	Integration
Energy (swing equation, single-node corpus)	186,060	1s timesteps
Tokamak (MHD)	167,500	envelope sweep
Atheric (Shannon/Friis)	15,830	packet-level
Terran (Boussinesq)	10,750	100 Hz Euler
Mycelial (Kirchhoff)	760	10 Hz network
Orbital (quaternion)	353	100 Hz Euler
Mars EDL (CO2 drag)	144	1000 Hz Euler
Plutonian (entropy decay)	71	100-yr timestep
Celestial (Yoshida)	71	3600s symplectic

Substrate	Rate (traj/s)	Integration
Swing Grid (intermittency envelope, multi-source)	7,777	envelope sweep
Marine (seawater)	11	1000 Hz Euler
Reactor (neutron kinetics)	5,578	envelope sweep
Asteroid (N-body GPU)	9	O(N^2) Metal

The compute spread between the fastest and slowest substrate: 20,673x. This is not a benchmark of hardware efficiency. It is a fingerprint of physics. The grid dynamics equation is simple. The O(N^2) rubble pile gravity is expensive. The engine's performance profile is determined entirely by the physical complexity of each domain.

No GPU cluster. No cloud infrastructure. No external physics license. The engine runs on one laptop and produces findings that no existing infrastructure has produced.

VII. Verifying a Finding

The verification surface for any finding in this document is the on-chain record at Internet Computer canister `ad7wi-4aaaa-aaaad-aeijq-cai`. The proof hashes cited throughout this document (every Master Hash, every per-environment hash) are entries in that immutable record. Anyone with access to the IC network can query the canister and confirm: this hash exists, it was sealed at this timestamp, it has not been modified.

Data products delivered to partners and customers (Section 08) carry per-trajectory or per-sweep proof hashes that match the on-chain anchor. An engineer in possession of a delivered data product can independently confirm three things: the data is what the engine produced, the engine produced it on the date of the on-chain seal, and the data has not been altered since.

The hash is the linkage. The chain is the witness. The engine produced both.

This document is the interface. The physics is the implementation. The proof is on chain.

What Is Available

Protocol Company / G^G Genesis Core Date: May 2026

The G^G corpus is not a research artifact. It is operating physics infrastructure, generating sealed trajectory and failure envelope data across thirteen substrates on a continuous basis. The data products below are the output of that infrastructure.

Energy Infrastructure

Reactor failure envelopes spanning the five-dimensional state space of neutron kinetics, thermal-hydraulics, and xenon-135 dynamics. Tokamak confinement envelopes mapping Z-axis shear and radial β-limit failure modes. Swing grid intermittency envelopes including the inertia-floor collapse maps. Each substrate is available as full envelope corpus or as targeted parameter slice with master proof hashes.

Orbital Systems

MAVEN-class attitude recovery trajectory corpus with full Euler rotational dynamics and quaternion propagation. GPS constellation maneuver kill chain envelopes with PDOP degradation maps by latitude band. EKF acceptance window envelopes characterizing chi-squared sensitivity to injection rate.

Drone Systems

EKF spoof acceptance envelopes across injection rate, PDOP, and velocity, including optimal stealth-rate identification. Friis topology severance envelopes covering asymmetric spoof, formation size, and frequency band selection. Beaufort energy envelopes mapping operational wind walls by formation size and load mass.

Humanoid Robotics

Bipedal gyroscopic gait corpus with full 3D Euler integration including the cross-product yaw coupling term. Yaw accumulation envelopes giving N_crit as a function of terrain friction, load mass, and load height. ZMP-blind regime maps identifying the parameter space where standard controllers cannot see the failure they are supposed to prevent.

Marine, Atheric, Mycelial, Terran, Mars EDL

Marine GPS-denied navigation corpus with 6-DOF hydrodynamics and IMU drift accumulation. Atheric clock-drift swarm collapse envelopes. Mycelial network resilience studies covering health-versus-density. Terran soil compaction envelopes with glomalin-coupled yield stress modeling across four soil types and eight robot body classes. Mars EDL ignition boundary corpus with full CO_2 atmospheric drag at 1000 Hz.

Custom Generation

The engine is configurable. New substrates, new parameter spaces, and targeted boundary investigations can be generated to specification. All output is sealed and reproducible.

Verification Stack

The SHA-256 proof chain architecture is available as a standalone delivery for organizations operating their own simulation infrastructure. Edge-deployable. No cloud dependency. Verification operates in denied, degraded, intermittent, and limited environments.

If your work touches any of these domains and you want the data, write to **kruze@aijesusbro.com**.

References

Citations are ordered by first appearance in the document. All equations and physical models reference their primary published sources.

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2. **Grad, H. and Rubin, H.** (1958). "Hydromagnetic equilibria and force-free fields." *Proceedings of the Second United Nations Conference on the Peaceful Uses of Atomic Energy*, Vol. 31. Grad-Shafranov equation for tokamak MHD equilibrium. Cited in Section 02 (plasma confinement physics).
3. **Anderson, P.M., Fouad, A.A.** (2003). *Power System Control and Stability*, 2nd Edition. IEEE Press. Swing equation for grid frequency dynamics. Cited in Section 02 (grid inertia and frequency stability).
4. **NVIDIA Corporation.** *Isaac Lab Documentation* (online). github.com/isaac-sim/IsaacLab. Accessed 2026. "Physics in IsaacLab are not deterministically reproducible when using env.reset, so demonstrations may fail on replay." Cited in Section 01 (non-determinism problem).
5. **U.S. Department of Defense.** (2020). *Global Positioning System Standard Positioning Service Performance Standard*, 5th Edition. GPS SPS 95th percentile PDOP bounds, Section 3.5. Cited in Section 03 (GPS geometry cascade).
6. **Boussinesq, J.** (1885). *Application des potentiels a l'etude de l'equilibre et du mouvement des solides elastiques*. Gauthier-Villars, Paris. Contact stress distribution equation for soil mechanics. Cited in Section 06 (terran soil contact).

Note on Friction Coefficients

Friction coefficient values used in Section 05 (wet mud, wet tile, dry gravel, dry concrete, rubber mat, dry asphalt) are drawn from standard tribology references for the corresponding surface materials. The values are within the commonly published ranges for each surface type.

Note on Proof Hashes

Every SHA-256 hash cited in this document is a primary reference. The hash is the evidence. On-chain records are queryable at canister `ad7wi-4aaaa-aaaad-aeijq-cai` on the Internet Computer. Verification operates against the on-chain anchor, independently of any infrastructure controlled by Protocol Company.

Coda

ChatGPT lands. A relationship I had been in ends in the same window. Someone in my professional circle spins up a WhatsApp mastermind to figure out what this thing is, and I jump in. That is how I enter AI. Through other people, building little projects together, learning by doing.

A few months later I realize that for the first time in my adult life I am unbound enough by physical responsibility to try something completely new. South America has been on my mind. I talk to my Peruvian family, people I have known since I was a kid, and the father offers his village. Come experience it, stay as long as you want. So I go. Laptop, no plan beyond the invitation. August 2023.

He is an elder in his village, a Quechua-speaking pueblo called Ccecca. He introduces me to the village as his son Jesús. That night I ask him about it. He tells me to let my old story go for a while, be his Peruvian son, and see what happens. So I do.

A few months later I travel to Pisac in the Sacred Valley to meet a friend I already know. I have long hair. I have been going by Hey Zeus. The friend asks what name I am using these days. Still Kruze? Hey Zeus, I tell him. Pisac decides on its own what comes next: the bro at the café who will not stop talking about agents, the hair, the Hey Zeus. AI Jesus. It sticks because the town named it. I keep it because the town named it.

I spend a year in the Sacred Valley taking AI contracts and riding every phase shift as it lands. People in the cafés start soliciting me for coaching. I run two-hour sessions walking them through how to implement AI into their strategies. Most of them have online businesses or businesses back in their home countries, and they pay me to help them learn AI naturally.

The rest of the people I meet in the cafés are uncomfortable with what is happening or are not sure what it is. What is it? Is it bad? That is the recurring question. I tell them the same thing. This is here. You have an opportunity to do things you could not do before. That has been my philosophical position from the start. Empowerment. AI as something an operator picks up to solve problems and understand things in ways that were not previously possible. AI native, AI first, since 2023. I run WhatsApp groups during this period telling everyone the same thing. The most important move right now is to spend time on these tools and go AI first.

That position is also why my simulation environments are MCP native. When MCP arrives, the rest of the world reads it as integration plumbing. I read it as substrate. The thing that lets an AI agent inhabit a synthetic environment. I decide my entire simulation stack should be MCP native so that I can watch Claude and Gemini fly spacecraft and walk into reactors. The environment has to be native to the AI for me to drive and for the agents to do. To do that I have to learn to code. I have never written a line. I learn through the agents themselves. The Rust workspace you have just read about in this document was built that way. I have still never written a line of code by hand.

The reason I built any of it is not defense. The reason is that I am interested in mycelial networks and the way trees coordinate without a central controller, in living systems that route around damage, in how machines and robots are going to engage with the planet and walk into the places humans cannot. Reactors. Deep sea. Asteroid debris fields. Contested electromagnetic environments. I built thirteen physics substrates because that is what the question requires if you want to answer it with math instead of marketing. The mycelial substrate in this document is not a quirk. It is the question the engine started from.

The work is for humanity. It is also for the AI. The abstraction layers AI has been trained on are not a fair representation of physical reality. Ground truth is medicine. This engine is one dose.

The defense applications are downstream of that question. So is the energy work. So is the orbital mechanics. The math turns out to apply to everything the autonomous systems industry is trying to build without math.

I have never written a line of code by hand. I built a deterministic physics engine across thirteen substrates by directing AI agents through MCP. I am the proof of my own thesis: that an individual paired with AI agents can produce work that organizations implementing AI structurally cannot. The math in this document is the receipts. This page is the reason.

John Kruze kruze@aijesusbro.com aijesusbro.com